

PRN : 202501120138

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3.1.1. Numpy Array Operations

01:21

Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using `ndim`.
- The shape of the array using `shape`.
- The total number of elements in the array using `size`.

Assume all input elements are valid integers.

Input Format:

- The first line contains two space-separated integers, `r` and `c`, representing the number of rows and the number of columns.
- The next `r` lines contain `c` space-separated integers representing the elements of the array, entered row by row.

Output Format:

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

Note: Use `reshape()` function to reshape the input array with the specified number of rows and columns.

Sample Test Cases

numpyarr...

Submit

1 `# import numpy as np`

2

3 `# write your code here...`

4 `import numpy as np`

5

6 `rows, cols = map(int, input().split())`

7

8

9 `elements = []`

10 `for _ in range(rows):`

11 `elements.extend(map(int, input().split()))`

Average time: 0.011 s, 10.92 ms

Maximum time: 0.021 s, 21.00 ms

2 out of 2 shown test case(s) passed

10 out of 10 hidden test case(s) passed

Test case 1: 21 ms

Expected output: 3 4, 1 2 3 4, 5 6 7 8, 9 10 11 12, [[1 2 3 4], [5 6 7 8]]

Actual output: 3 4, 1 2 3 4, 5 6 7 8, 9 10 11 12, [[1 2 3 4], [5 6 7 8]]

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3.2.1. Numpy: Matrix Operations

03:19

The given code takes two 3×3 matrices, `matrix_a`, and `matrix_b`, as input from the user and converts them into NumPy arrays.

Task:

You are required to compute and display the results of the following matrix operations:

- Addition (`matrix_a + matrix_b`)
- Subtraction (`matrix_a - matrix_b`)
- Element-wise Multiplication (`matrix_a * matrix_b`)
- Matrix Multiplication (`matrix_a . matrix_b`)
- Transpose of Matrix A

Input Format:

- The user will input 3 rows for `matrix_a`, each containing 3 integers separated by spaces.
- Similarly, the user will input 3 rows for `matrix_b`, each containing 3 integers separated by spaces.

Output Format:

The program should display the results of the operations in the following order:

- The result of Addition.
- The result of Subtraction.
- The result of Element-wise Multiplication.
- The result of Matrix Multiplication.
- The Transpose of Matrix A.

matrixOp...

Submit

1 `# import numpy as np`

2

3 `## Input matrices`

4 `# print("Enter Matrix A:")`

5 `# matrix_a = np.array([list(map(int, input().split())) for i in range(3)])`

6

7 `# print("Enter Matrix B:")`

8 `# matrix_b = np.array([list(map(int, input().split())) for i in range(3)])`

9

Average time: 0.018 s, 18.80 ms

Maximum time: 0.024 s, 24.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1: 22 ms

Expected output: Enter Matrix A: 1 2 3, 4 5 6, 7 8 9, Enter Matrix B: 1 1 1

Actual output: Enter Matrix A: 1 2 3, 4 5 6, 7 8 9, Enter Matrix B: 1 1 1

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3.2.2. Numpy: Horizontal and Vertical Stacking of Arrays

02:33

You are given two arrays `arr1` and `arr2`. You need to perform horizontal and vertical stacking operations on them using NumPy.

- **Horizontal Stacking:** Stack the two matrices horizontally (side by side).
- **Vertical Stacking:** Stack the two matrices vertically (one below the other).

Input Format:

- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

Output Format:

- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

Sample Test Cases

stacking.py

Submit

1 import numpy as np

2

3 # Input matrices

4 print("Enter Array1:")

5 arr1 = np.array([list(map(int, input().split())) for i in range(3)])

6

7 print("Enter Array2:")

8 arr2 = np.array([list(map(int, input().split())) for i in range(3)])

9

10 # Perform horizontal stacking (hstack)

11 horizontal=np.hstack((arr1,arr2))

Average time 0.014 s 14.00 ms Maximum time 0.015 s 15.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 15 ms Debug

Expected output

Enter Array1:

1 2 3

4 5 6

7 8 9

Enter Array2:

4 5 6

Actual output

Enter Array1:

1 2 3

4 5 6

7 8 9

Enter Array2:

4 5 6

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3.2.3. Numpy: Custom Sequence Generation

06:33

Write a Python program that takes the following inputs from the user:

- Start value: The starting point of the sequence.
- Stop value: The sequence should end before this value.
- Step value: The increment between each number in the sequence.

The program should then generate a sequence using `numpy` based on these inputs and print the generated sequence.

Input Format:

- The user will input three integer values: start, stop, and step, each on a new line.

Output Format:

- The program should print the generated sequence based on the input values.

Sample Test Cases

customS...

Submit

1 import numpy as np

2

3 # Take user input for the start, stop, and step of the sequence

4 start = int(input())

5 stop = int(input())

6 step = int(input())

7

8 # Generate the sequence using np.arange()

9 seq=np.arange(start,stop,step)

10 # Print the generated sequence

11 print(seq)

Average time 0.007 s 6.75 ms Maximum time 0.010 s 10.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 10 ms Debug

Expected output

3

10

2

[3 5 7 9]

Actual output

3

10

2

[3 5 7 9]

Test case 2 7 ms

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3.2.4. Numpy: Arithmetic and Statistical Operations, Mathematical Operations, Bitw... 04.03

You are given two arrays A and B. Your task is to complete the function array_operations, which will convert these lists into NumPy arrays and perform the following operations:

1. Arithmetic Operations:
• Compute the element-wise sum, difference, and product of the two arrays.
2. Statistical Operations:
• Calculate the mean, median, and standard deviation of array A.
3. Bitwise Operations:
• Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: A_i OR B_i).

Input Format:
• The first line contains space-separated integers representing the elements of array A.
• The second line contains space-separated integers representing the elements of array B.

Output Format:
• For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

Sample Test Cases

different...

1 import numpy as np
2
3 def array_operations(A, B):
4
5 # Convert A and B to NumPy arrays
6 A = np.array(A)
7 B = np.array(B)
8 # Arithmetic Operations
9 sum_result = A+B
10 diff_result = A-B
11 prod_result = A*B

Average time 0.008 s 8.00 ms Maximum time 0.011 s 11.00 ms 1 out of 1 shown test case(s) passed 2 out of 2 hidden test case(s) passed

Test case 1 11 ms Debug

Expected output
1 2 3 4
5 6 7 8
Element-wise Sum: 6 8 10 12
Element-wise Difference: -4 -4 -4 -4
Element-wise Product: 5 12 21 32
Mean of A: 2.5

Actual output
1 2 3 4
5 6 7 8
Element-wise Sum: 6 8 10 12
Element-wise Difference: -4 -4 -4 -4
Element-wise Product: 5 12 21 32
Mean of A: 2.5

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3.2.5. Numpy: Copying and Viewing Arrays 05.07

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:
• Creating a view of the original_array and assigning it to view_array.
• Creating a copy of the original_array and assigning it to copy_array.

After completing these steps, observe how modifying the view affects the original_array, while modifying the copy does not.

Input Format:
• A single line of space-separated integers.

Output Format:
• After modifying the view:
Original array after modifying view: <original_array>
View array: <view_array>

• After modifying the copy:
Original array after modifying copy: <original_array>
Copy array: <copy_array>

Sample Test Cases

copyAnd...

1 import numpy as np
2
3 inputlist = list(map(int,input().split(" ")))
4
5 # Original array
6 original_array = np.array(inputlist)
7
8 # Create a view
9 view_array =original_array.view()
10
11 # Create a copy

Average time 0.004 s 4.60 ms Maximum time 0.007 s 7.00 ms 2 out of 2 shown test case(s) passed 2 out of 2 hidden test case(s) passed

Test case 1 7 ms Debug

Expected output
10 20 30 40 50 60 70 80
Original array after modifying view: [99 20 30 40 50 60 70 80]
View array: [99 20 30 40 50 60 70 80]
Original array after modifying copy: [99 20 30 40 50 60 70 80]
Copy array: [10 88 30 40 50 60 70 80]

Actual output
10 20 30 40 50 60 70 80
Original array after modifying view: [99 20 30 40 50 60 70 80]
View array: [99 20 30 40 50 60 70 80]
Original array after modifying copy: [99 20 30 40 50 60 70 80]
Copy array: [10 88 30 40 50 60 70 80]

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3.2.6. Numpy: Searching, Sorting, Counting, Broadcasting

05.23

The given code in the editor takes a single array, array1, as space-separated integers as input from the user.
Additionally, it takes the following inputs:

- search_value: The value to search for in the array.
- count_value: The value to count its occurrences in the array.
- broadcast_value: The value to add for broadcasting across the array.

You need to complete the code to perform the following operations:

- Searching:** Find the indices where search_value appears in array1 and print these indices.
- Counting:** Count how many times count_value appears in array1 and print the count.
- Broadcasting:** Add broadcast_value to each element of array1 using broadcasting, and print the resulting array.
- Sorting:** Sort array1 in ascending order and print the sorted array.

Input Format:

- A single line containing space-separated integers representing array1.
- An integer search_value represents the value to search for in the array.
- An integer count_value represents the value to count in the array.
- An integer broadcast_value represents the value to add to each element of the array.

Output Format:

- The indices where search_value occurs in array1.
- The count of occurrences of count_value in array1.
- The array after adding the broadcast_value to each element.

arrayOpe...

Submit

1 import numpy as np

2

3 # Input array from the user

4 array1 = np.array(list(map(int, input().split())))

5

6 # Searching

7 search_value = int(input("Value to search: "))

8 count_value = int(input("Value to count: "))

9 broadcast_value = int(input("Value to add: "))

10

11 # Find indices where value matches in array1

Average time 0.011 s 10.50 ms Maximum time 0.012 s 12.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 12 ms Debug

Expected output 1 1 1 2 2 2

Actual output 1 1 1 2 2 2

Value to search: 1

Value to count: 2

Value to add: 2

[0 1 2]

3

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3.2.7. Student Data Analysis and Operations

15.55

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

- Print all student details: Display the complete details of all students, including roll numbers and marks for all subjects.
- Find total students: Determine the total number of students in the dataset.
- Print all student roll numbers: Extract and print the roll numbers of all students.
- Print Subject 1 marks: Extract and print the marks of all students in Subject 1.
- Find minimum marks in Subject 2: Identify the lowest marks in Subject 2.
- Find maximum marks in Subject 3: Identify the highest marks in Subject 3.
- Print all subject marks: Display the marks of all students for each subject.
- Find total marks of students: Compute the total marks for each student across all subjects.
- Find the average marks of each student: Compute the average marks for each student.
- Find average marks of each subject: Compute the average marks for all students in each subject.
- Find average marks of Subject 1 and Subject 2: Compute the average marks for Subject 1 and Subject 2.
- Find average marks of Subject 1 and Subject 3: Compute the average marks for Subject 1 and Subject 3.
- Find the roll number of the student with maximum marks in Subject 3: Identify the student with the highest marks in Subject 3 and print their roll number.
- Find the roll number of the student with minimum marks in Subject 2: Identify the student with the lowest marks in Subject 2 and print their roll number.
- Find the roll number of students who scored 24 marks in Subject 2: Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.
- Find the count of students who got less than 40 marks in Subject 1: Count the number of

Operatio...

Submit

1 import numpy as np

2

3 a = np.loadtxt("Sample.csv", delimiter=',', skiprows=1)

4

5 import numpy as np

6

7 a = np.loadtxt("Sample.csv", delimiter=',', skiprows=1)

8

9 # 1. Print all student details

10 print("All student Details:\n",a)

11

Average time 0.013 s 13.00 ms Maximum time 0.013 s 13.00 ms

1 out of 1 shown test case(s) passed

Test case 1 13 ms Debug

Expected output All student Details:
[[301, 67, 77, 88.]
[302, 78, 88, 77.]
[303, 45, 56, 89.]
[304, 88, 98, 45.]
[305, 78, 88, 99.]

Actual output All student Details:
[[301, 67, 77, 88.]
[302, 78, 88, 77.]
[303, 45, 56, 89.]
[304, 88, 98, 45.]
[305, 78, 88, 99.]

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